Chapter 5 - Vector Calculus

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Exercise 5.1:

Compute the derivative $f'(x)$ for

$$f(x) = log(x^4)sin(x^3)$$

Solution.

recall the product rule, which is

$$\frac{d}{dx}[f(x)g(x)] = f(x)g'(x) + f'(x)g(x)$$

with $f(x)$ as $log(x^4)$ and $g(x)$ as $sin(x^3)$

we can get something like this

$$\frac{d}{dx}[f(x)] = [log(x^4)\frac{d}{dx}[sin(x^3)]] + [\frac{d}{dx}[log(x^4)]sin(x^3)]$$

then we can define each $\frac{d}{dx}$s by applying chain rule

$$\frac{d}{dx}[sin(x^3)] = 3x^2cos(x^3)$$

$$\frac{d}{dx}[log(x^4)] = \frac{1}{x^4}[4x^3]$$
$$\frac{d}{dx}[log(x^4)] = \frac{4}{x}$$

well, we can finally get our final solution:

$$f'(x) = 3x^2cos(x^3)log(x^4) + \frac{4}{x}sin(x^3)$$

Exercise 5.2:

Compute the derivative $f'(x)$ of the logistic sigmoid

$$f(x) = \frac{1}{1 + \exp(-x)}.$$

Solution.

we can write it as

$$f(x) = [1 + \exp(-x)]^{-1}$$

let $u(x) = [1 + \exp(-x)]$

$$\frac{d}{dx}[f(x)] = \frac{d}{dx}[u(x)]^{-1}$$
$$\frac{d}{dx}[f(x)] = -1 * u(x)^{-2} * u'(x)$$

refer to the sum rule, we get $u'(x) = -1 * \exp(-x)$

then we can substitute back to the chain rule we defined above

$$\frac{d}{dx}[f(x)] = -1 * [1 + \exp(-x)]^{-2} * -1 * \exp(-x)$$
$$\frac{d}{dx}[f(x)] = [1 + \exp(-x)]^{-2} * \exp(-x)$$

here we get the final answer of the derivative of sigmoid func

$$f'(x) = \frac{\exp(-x)}{[1 + \exp(-x)]^2}$$

also the fun part of this thing is we can evaluate that

$$\sigma(x) = \frac{1}{1 + \exp(-x)}.$$
$$1 - \sigma(x) = 1 - \frac{1}{1 + \exp(-x)}.$$
$$1 - \sigma(x) = \frac{exp(-x)}{1 + \exp(-x)}.$$

which means we can do this

$$\sigma(x) * (1 - \sigma(x)) = \frac{1}{1 + \exp(-x)} * \frac{exp(-x)}{1 + \exp(-x)}$$
$$f'(x) = \sigma(x) * (1 - \sigma(x)) = \frac{\exp(-x)}{[1 + \exp(-x)]^2}$$

we can actually use this identity as some kind of 'fast path' in backpropagation in the low-level machine learning calculation, so that we dont have to calculate the sigmoid func multiple times, we can just initiate it once, then we should definitely use the identity for computation efficiency.

Exercise 5.3:

Compute the derivative $f'(x)$ of the function

$$f(x) = \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right),$$

where $\mu, \sigma \in \mathbb{R}$ are constants.

Solution.

welp, from here we can just rinse and repeat our progress before, as usual, refer to the chain rule

let

$$u(x) = -\frac{1}{2\sigma^2}(x-\mu)^2$$

means that

$$\frac{d}{dx}[f(x)] = exp(u(x))\frac{d}{dx}[u(x)]$$

we can conclude

$$\frac{d}{dx}[u(x)] = -\frac{(x-\mu)}{\sigma^2}$$

finally, lemon squeezy

$$f'(x) = -\frac{(x-\mu)}{\sigma^2}exp(-\frac{1}{2\sigma^2}(x-\mu)^2)$$

Exercise 5.5:

Consider the following functions:

$$f_1(\boldsymbol{x}) = \sin(x_1)\cos(x_2)\,, \quad \boldsymbol{x} \in \mathbb{R}^2$$
$$f_2(\boldsymbol{x}, \boldsymbol{y}) = \boldsymbol{x}^\top \boldsymbol{y}\,, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$$
$$f_3(\boldsymbol{x}) = \boldsymbol{x}\boldsymbol{x}^\top\,, \quad \boldsymbol{x} \in \mathbb{R}^n$$

a. What are the dimensions of $\frac{\partial f_i}{\partial \boldsymbol{x}}$?

b. Compute the Jacobians.

Solution.

we can simply comprehend jacobians as like a way to measure what kind of output that we will get out of a certain equation, imagine that pytorch / tensorflow is just an optimized engine to calculate the chain rule (while updating its own weights in backprop), those chain rules are just jacobian matrices multiplication, it's just matmul all the way to the down :)

well the first thing we need to consider in order to determine the dimensions of these $\frac{\partial f}{\partial \boldsymbol{x}}$ is to look at the input dimensions of x, the product of equations and combine them

case f_1:
we can see that it takes x_1 and x_2 or just simply look at the notation $\quad \boldsymbol{x} \in \mathbb{R}^2$, well basically it means the same thing

yea so we can deduct the result to be some kind of row matrix (1 x 2) matrix

so what about the jacobian?

just do partial differential on the equation

$$f_1(\boldsymbol{x}) = \sin(x_1)\cos(x_2)\,, \quad \boldsymbol{x} \in \mathbb{R}^2$$

$$\frac{\partial f_1}{\partial x_1} = \cos(x_1)\cos(x_2)$$
$$\frac{\partial f_1}{\partial x_2} = -\sin(x_1)\sin(x_2)$$

we can conclude that the jacobian is

$$[\cos(x_1)\cos(x_2), -\sin(x_1)\sin(x_2)]$$

case f_2:

rinse and repeat:

$$f_2(\boldsymbol{x}, \boldsymbol{y}) = \boldsymbol{x}^\top \boldsymbol{y}\,, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$$

$$f_2(\boldsymbol{x}, \boldsymbol{y}) = \boldsymbol{x}^\top \boldsymbol{y} = x_1y_1 + x_2y_2 + \cdots + x_ny_n\,, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$$

taking the partial derivative with respect to any single component x_i, treating y_i as a constant:

$$\frac{\partial f_2}{\partial x_i} = y_i$$

we can conclude that the Jacobian is a $1 \times n$ row vector:

$$\left[\frac{\partial f_2}{\partial x_1}, \frac{\partial f_2}{\partial x_2}, \dots, \frac{\partial f_2}{\partial x_n}\right] = [y_1, y_2, \dots, y_n] = \boldsymbol{y}^\mathsf{T}$$

case f_3:

here, multiplying an $n \times 1$ vector by a $1 \times n$ vector outputs an $n \times n$ matrix:

$$f_3(\boldsymbol{x}) = \boldsymbol{x}\boldsymbol{x}^\mathsf{T}, \quad \boldsymbol{x} \in \mathbb{R}^n$$

let's look at a single element of this resulting matrix at row i and column j, which is simply $x_i x_j$. We take the partial derivative of this element with respect to an input variable x_k using the product rule:

$$\frac{\partial(x_i x_j)}{\partial x_k} = \frac{\partial x_i}{\partial x_k} x_j + x_i \frac{\partial x_j}{\partial x_k}$$

using the Kronecker delta ($\delta_{ik} = 1$ if $i = k$, and 0 otherwise) to represent the derivatives of the individual variables:

$$\frac{\partial(x_i x_j)}{\partial x_k} = \delta_{ik} x_j + x_i \delta_{jk}$$

we can conclude that the Jacobian forms an $n \times n \times n$ 3D tensor, where each element is defined as:

$$(J_3)_{i,j,k} = \delta_{ik} x_j + x_i \delta_{jk}$$

Exercise 5.6:

Differentiate f with respect to t and g with respect to X, where

$$\begin{aligned} f(t) &= \sin(\log(t^\mathsf{T} t))\,, \quad t \in \mathbb{R}^D \\ g(X) &= \operatorname{tr}(AXB)\,, \quad A \in \mathbb{R}^{D\times E}, X \in \mathbb{R}^{E\times F}, B \in \mathbb{R}^{F\times D}\,, \end{aligned}$$

where $\operatorname{tr}(\cdot)$ denotes the trace.

Solution.

Exercise 5.7:

Compute the derivatives $df/d\boldsymbol{x}$ of the following functions by using the chain rule. Provide the dimensions of every single partial derivative. Describe your steps in detail.

a.

$$f(z) = \log(1+z)\,, \quad z = \boldsymbol{x}^\top \boldsymbol{x}\,, \quad \boldsymbol{x} \in \mathbb{R}^D$$

b.

$$f(z) = \sin(z)\,, \quad z = A\boldsymbol{x} + b\,, \quad A \in \mathbb{R}^{E\times D}, \boldsymbol{x} \in \mathbb{R}^D, b \in \mathbb{R}^E$$

where $\sin(\cdot)$ is applied to every element of z.

Solution.

Exercise 5.8:

Compute the derivatives $df/d\boldsymbol{x}$ of the following functions. Describe your steps in detail.

a. Use the chain rule. Provide the dimensions of every single partial derivative.

$$\begin{aligned} f(z) &= \exp(-\frac{1}{2}z) \\ z = g(\boldsymbol{y}) &= \boldsymbol{y}^\top S^{-1}\boldsymbol{y} \\ \boldsymbol{y} = h(\boldsymbol{x}) &= \boldsymbol{x} - \mu \end{aligned}$$

where $\boldsymbol{x}, \mu \in \mathbb{R}^D$, $S \in \mathbb{R}^{D\times D}$.

b.
$$f(\boldsymbol{x}) = \mathrm{tr}(\boldsymbol{x}\boldsymbol{x}^\top + \sigma^2 I)\,, \quad \boldsymbol{x} \in \mathbb{R}^D$$

Here $\mathrm{tr}(A)$ is the trace of A, i.e., the sum of the diagonal elements A_{ii}.
Hint: Explicitly write out the outer product.

c. Use the chain rule. Provide the dimensions of every single partial derivative. You do not need to compute the product of the partial derivatives explicitly.

$$\begin{aligned} f &= \tanh(z) \in \mathbb{R}^M \\ z &= A\boldsymbol{x} + b\,, \quad \boldsymbol{x} \in \mathbb{R}^N, A \in \mathbb{R}^{M\times N}, b \in \mathbb{R}^M. \end{aligned}$$

Here, tanh is applied to every component of z.

Solution.

Exercise 5.9:

We define

$$
\begin{aligned}
g(\boldsymbol{x}, z, \nu) &:= \log p(\boldsymbol{x}, z) - \log q(z, \nu)\\
z &:= t(\epsilon, \nu)
\end{aligned}
$$

for differentiable functions p, q, t and $\boldsymbol{x} \in \mathbb{R}^D, z \in \mathbb{R}^E, \nu \in \mathbb{R}^F, \epsilon \in \mathbb{R}^G$. By using the chain rule, compute the gradient

$$\frac{d}{d\nu} g(\boldsymbol{x}, z, \nu).$$

Solution.